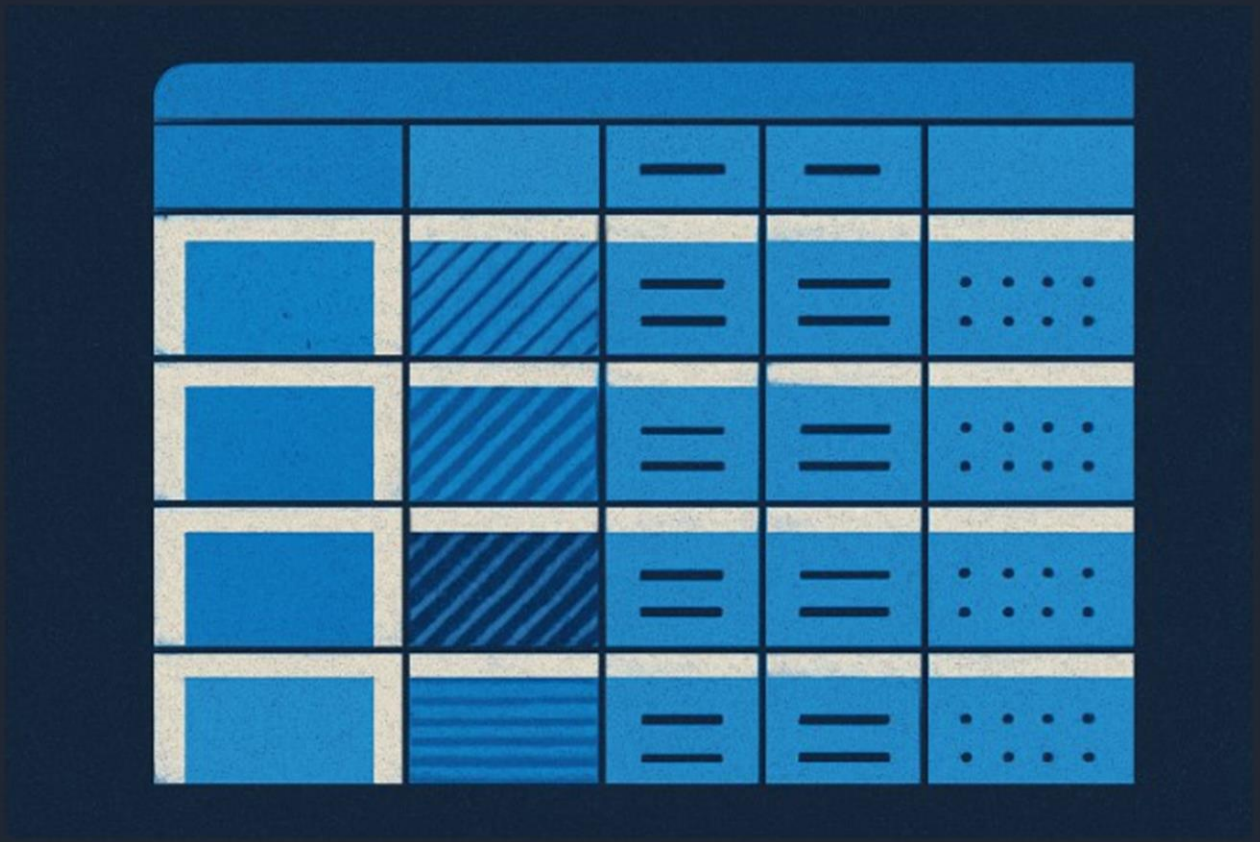
Customer Behaviour Analysis

Complete Data Analysis Project

EXCEL PYTHON MYSQL POWER BI

About the Dataset



Used for **customer behavior analysis** and business decision-making — tracking purchases, spend, frequency, and discount response.

Type	Level	Format	Data Mix
Retail / E-commerce	Customer Transaction	Structured Tabular	Numerical + Categorical

Why This Data Matters

High-Value Customers

Identify top spenders

Purchase Patterns

Understand buying behavior

Discount Strategy

Optimize promotions

Marketing ROI

Improve campaigns

Customer Lifetime Value

Increase CLV

Business Problem

> How can a retail company understand customer purchasing behavior to increase revenue, improve retention, and optimize marketing strategies?

- **Low Retention**
  Customers not purchasing frequently
- **Ineffective Campaigns**
  Discounts may not drive purchases
- **Product Demand**
  Which categories perform best?
- **Payment Behavior**
  Are digital payments increasing?
- **Seasonal Variation**
  Which season drives maximum purchases?
- **Customer Segmentation**
  High-value vs. discount-sensitive buyers

Data Dictionary

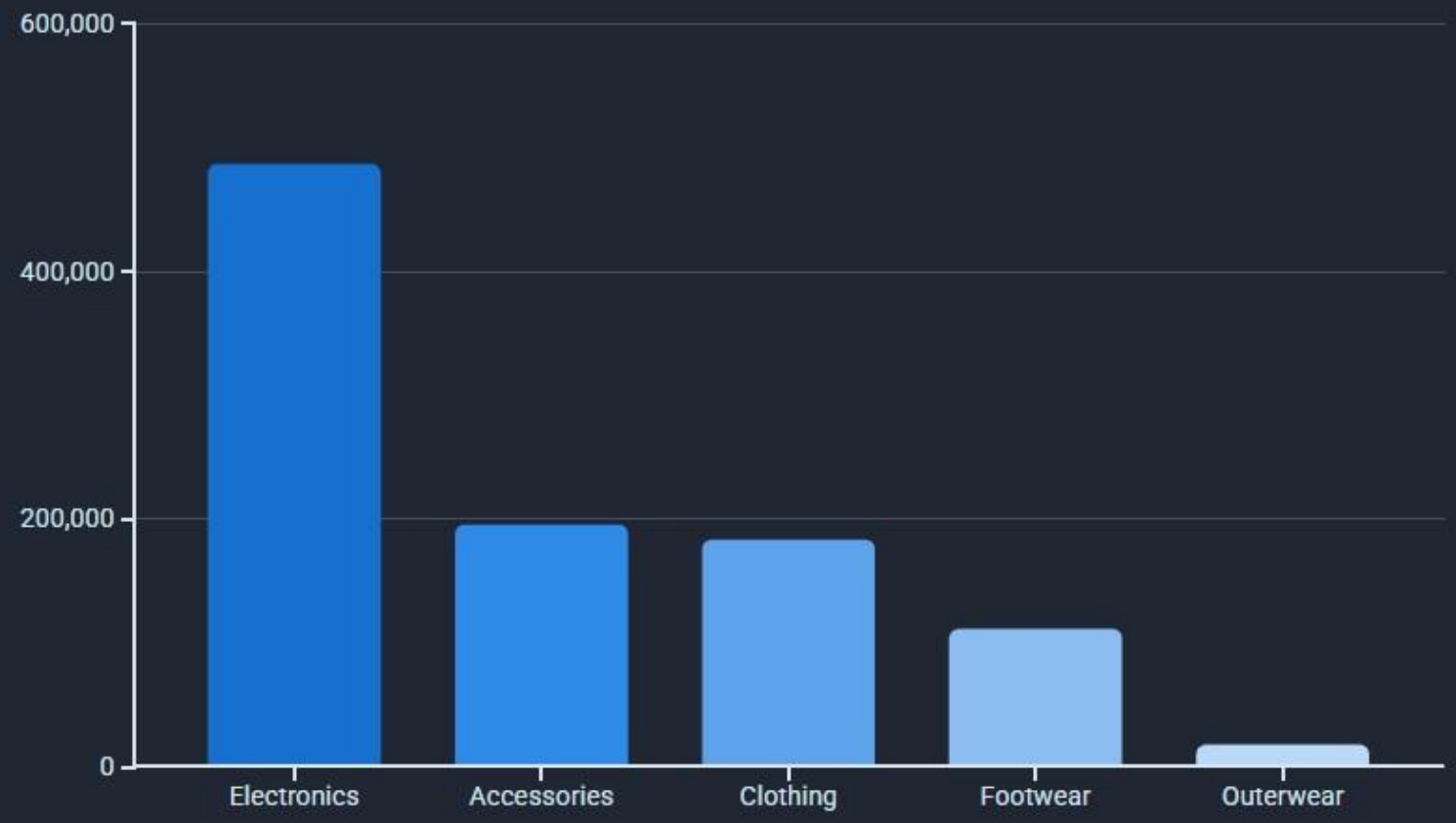
Customer ID	Unique identifier
Age / Gender	Demographics
Category / Item	Product details
Purchase Amount	Amount spent
Location	Geographic region
Season	Winter, Summer, etc.
Review Rating	Customer rating
Subscription Status	Yes / No
Discount Applied	Promo code usage
Previous Purchases	Past purchase count
Payment Method	Mode of payment
Frequency	Purchase frequency

EDA — Data Cleaning (Python)

Key steps: category correction via mapping dictionary, category-specific missing value handling (e.g., Electronics → "Not Applicable"), product-level mean for Review Rating, zero-fill for Previous Purchases, duplicate removal by Customer ID, and column standardization for SQL/Power BI compatibility.

SQL Analysis — Revenue Insights

Revenue by Category



Discount Impact

With discount: **$219** avg purchase
Without: **$182**

Subscribers Spend More

Yes: **$270** avg
No: **$166** avg

Top Age Group

51+ generates **$399K** — highest revenue

SQL Analysis — Customer Segments

Repeat Buyers & Subscription

Repeat buyers (5+ purchases) show strong subscription correlation — **1,190** subscribed vs. **2,763** not subscribed.

Top Products by Category

- Clothing: Shirt (315), Blouse (171), Pants (171)
- Electronics: Phone (171), Headphones (166), Watch (157)
- Footwear: Shoes (303), Sandals (160), Sneakers (145)
- Accessories: Jewelry (171), Bag (164), Belt (161)

Power BI Dashboard

5K	30	3.62
Total Customers	Unique Items	Avg Review Rating

$994K	$199
Total Spend	Avg Spend

Key visuals: Revenue by Gender (Female 39.33%, Other 35.41%, Male 25.26%), Category (Electronics $0.49M leads), Season (Spring & Winter $0.24M each), Age Group (26–35 leads at 40.17%), Payment Method (Debit Card $0.25M top), and Location (Chicago avg $707, New York total $187K).

Key Take Away

Electronics Dominates

49% of total revenue — focus inventory here

Discounts Raise Order Value

Discounted orders average **$219** vs. $182

Subscribers Are High-Value

Subscribers spend **63% more** on average

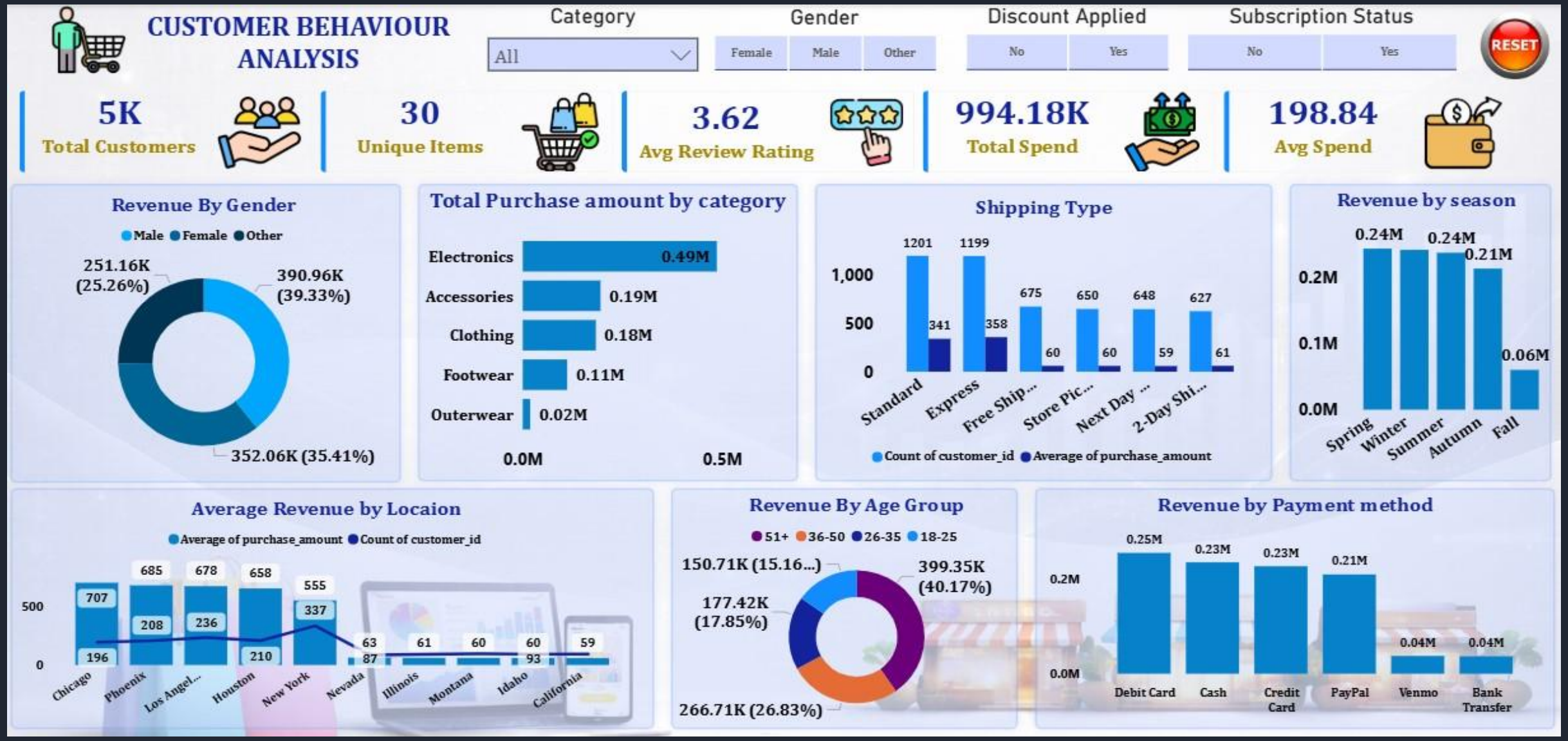
Loyal Customers = 62%

3,085 of 5,000 are repeat buyers

Age 26–35 Drives Revenue

Highest segment at **40%** of total spend

Customer Behavior Revenue



Best vs Worst Analysis